DSA ki Baithak

CODE BAITHAK

BY NAMAN BHAIYA

Recap (Previous Lecture Connection):

Number System :)

1. Sum Of Digits

Input :

$n = 687$ $\quad$ $\%10$ $\quad$ 6, 8, 7 → sum

Output :

21 $\quad$ $6 + 8 + 7$

① Problem Read

②

③ Pseudocode

$n = 687$ $\quad$ 7

```
while(n>0){
    digit = n%10
    sum += digit
    n/=10
}
```

④ Code

$n = \cancel{687}\ \cancel{68}\ \cancel{6}\ 0$

$sum = 0 + 7 + 8 + 6$

$digit = \cancel{7}\ \cancel{8}\ 6$

$sum = 21$

```
while (n > 0) {
    int digit = n % 10
    n /= 10
    sum += digit
}
```

2. Add Digits

Input :

num = 38

Output :

2

$3+8 = 11$

$\downarrow$

$1+1 = 2$

[0, 9]

1

2 num = sum → (num > 9)

10, 11, 12

3 while (num > 9)

sum

num = sum

4 Code

num = 38 11 2

num → 2

Sum = 11 2

num

```
while (num > 9) {
    int sum = 0
    int n = num
    while (n > 0) {
        int digit = n%10
        n/= 10
        sum += digit
    }
    num = sum
}
```

3. Reverse Digits

Input :

$n = 12345 \rightarrow$

Output :

54321

54321

5 4

5x10

% 10

/ 10

①

②

50 + 3 → 53

5 → 54 → 50 + 4

54 → 50 + 4

= 54

54 → 3

540 + 3 → 543

```
while (n > 0)
    digit = n % 10
    num = num x 10 + digit
    n /= 10
```

③ Pseu

④ Code

$n = \cancel{123} \xrightarrow{\div 10} 12 \xrightarrow{\div 10} 0$

321

$num = \cancel{0 \times 10 + 3} \; \cancel{3 \times 10 + 2}$

$\cancel{32 \times 10 + 1}$

321

digit = ~~3~~ ~~2~~ 1

num → 321

123

```
num = 0
while (n > 0) {
    int digit = n % 10
    n /= 10
    num = num * 10
          + digit
}
```

4. Palindrome Number

$\boxed{\text{Org} = \text{Rever}}$

Naman

$n = 123$ → 123, 321 ✗

false

Input:

$n = 121$ → 121, 121

Output:

true

1. ✓
2. Reversed num == num → true
3.
4. Code

5. Count Digits that Divide

2 4 6 9

✓ ✗ ✓ ?

2,3

Input :

$n = 9642$

Output :

2

1

(n % digit == 0)

Count ++

2

3

4 Code

6. Armstrong Number

Input :

$n = 153$

countDigits = 3

$1^3 + 5^3 + 3^3 =$ org num

Output :

true

$[1^3 + 5^3 + 3^3 = 153]$

1

2 Sum += Math.pow(digit, cd)

3

4 Code

$n = 153$

total_digits = 3

$sum = 3^3 + 5^3 + 1^3$

sum == org

true

```
org
~~temp~~ = n
sum = 0
total_digits = countDigits(n)
while (n > 0) {
    rem = n % 10
    sum += (int) Math.pow(rem, digits)
    n /= 10
}
return sum == num
```

7. Automorphic Number

① ✓

n = 123 → 3

Square = n*n

② Countdigits →

123 % 10 = 3

123 % 100 = 23

123 % 1000 = 123

Input :

n = 76 → 2

Output :

org = = Square % Math.pow(10, total)

→ true

Automorphic

[76^2 = 5776]

$n = 76$

$Square = \underline{76 * 76}$

$total_digits = 2$

$last = \underline{5776} \% \underline{Math.pow(10, 2)}$

$\downarrow$

$\underline{100}$

76

$n == last$

$Square = n * n$

$digits = countDigits(n)$

$n == square \% (int)\ Math.pow(10, digits)$

8. Perfect Number

Input :

$n = 145$

Output :

1

$[1! + 4! + 5! = 145]$